

Fallou Diouf

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[GitHub](#) | [Portfolio](#) | [LinkedIn](#)

PROFESSIONAL OBJECTIVE

I am seeking a **4 to 6-month final-year internship** starting in **February 2027**, in the fields of **Artificial Intelligence, Computer Vision, and Data Science**.

EDUCATION

2025 – 2027 | Université Paris Cité, Paris, France

Master's Degree in Computer Science – Vision and Machine Intelligence *Key modules:* Programming (Python, C++), Computer Vision, Image Analysis, 3D and Biomedical Imaging, Signal and Image Processing, Vision System Modeling, Deep Learning, Pattern Recognition, Data Engineering (Data Science, Decision Support), Applied Mathematics, and Research Methodology.

2023 – 2024 | Université Gaston Berger, Saint-Louis, Senegal

Master 1 in Computer Science – Data Management and Software Engineering (GDIL) *Key modules:* Artificial Intelligence, Advanced Algorithms, Complexity and Information Theory, Languages, Automata and Compilation, Advanced Programming (C/C++), Object-Oriented Modeling (UML), Advanced Databases (SQL), Applied Mathematics, XML & Web Services.

2020 – 2023 | Université Gaston Berger, Saint-Louis, Senegal

Bachelor's Degree in Computer Science – Information Systems *Key modules:* Algorithms and Programming, Object-Oriented Programming, Data Structures, Mathematics (Probability, Statistics, Algebra), UML Analysis and Design, Software Engineering, Data Modeling, Relational Databases, Information Systems Architecture.

PROFESSIONAL EXPERIENCE

Programming Tutor – C/C++, Algorithms, OOP and SQL

2023 – 2024

- Re-explained complex concepts from lectures, tutorials and practical sessions, adapting explanations to individual difficulties.
- Monitored learners' progress and supported them in developing greater autonomy.

ACADEMIC PROJECTS

2025 – 2026 | Person Tracking with YOLOv8 and DeepSORT

Individual

Development of a system for detecting and tracking people in videos using YOLOv8 and DeepSORT. Assignment of unique identifiers (*Track ID*) and analysis of people's trajectories over time.

Skills: Computer Vision, Object Detection, Object Tracking, YOLOv8, DeepSORT, OpenCV.

2025 – 2026 | Coin Detection and Classification

Team & Individual

Classical Image Processing (team): Design of a pipeline using Gaussian filtering, Canny edge detection and the Hough Transform to segment coins, combined with texture and diameter analysis methods to automate coin classification and total amount calculation.

Deep Learning (individual): Development of an AlexNet-based classifier to identify the origin and value of coins, with optimization of feature extraction.

Skills: Computer Vision, Classification, OpenCV, PyTorch, Transfer Learning, Comparative Evaluation.

2025 – 2026 | Self-Supervised Test-Time Training

Pair Project

Implementation of a *Self-Supervised Learning* approach combining SimCLR and Vision Transformer to improve the robustness of an image classification model during inference.

Validation accuracy of **90.08%** and an improvement of **+0.39 percentage points** on CIFAR-10-C with Test-Time Training.

Skills: PyTorch, SimCLR, Vision Transformer, Self-Supervised Learning, Test-Time Training.

2025 – 2026 | Information Retrieval Engine

Team Project

Design of a hybrid search engine combining lexical retrieval (BM25) and semantic search (Embeddings) for multi-domain queries.

Development of a *Reranking* module and integration of LangChain/LangGraph to improve Top-K result relevance. Best result with embeddings: **70.76% Recall@20**, **52.02% MRR** and **30.6 ms** latency.

Skills: Information Retrieval, RAG, LLM, Embeddings, LangChain, Performance Evaluation.

TECHNICAL SKILLS

Computer Vision & Image Processing Object Detection, Object Tracking, Image Classification, Image Segmentation, Feature Extraction, Shape Recognition, Video Analysis, OpenCV, YOLOv8, DeepSORT.

Deep Learning & Machine Learning PyTorch, TensorFlow, CNN, AlexNet, ResNet, DenseNet, Vision Transformer, Transfer Learning, Self-Supervised Learning, SimCLR, Test-Time Training, Scikit-learn.

Information Retrieval & NLP TF-IDF, BM25, Embeddings, BGE, RAG, LLM, LangChain, LangGraph, Reranking, Semantic Search.

Programming, Data & Tools Python, C, C++, JavaScript, SQL, NumPy, Pandas, PostgreSQL, Git, GitHub, Docker, Linux, FastAPI, Django, PySpark.

LANGUAGES

French: Fluent

English: Intermediate

INTERESTS

Artificial Intelligence • Scientific Research • Technology Trends • Programming